等离子体波

线性，小振幅：

λ、ω、ϕ、$\vec{k}$、偏振.

偏振：线偏，圆偏，椭偏.

（E的矢量变化方向）

相位. $E = E_0 \exp(i\phi)$. $\phi = \underset{\Delta}{\omega t} - \underset{\Delta}{\vec{k}\cdot\vec{r}}$.

相速度. $\phi = \omega t - \vec{k}\cdot\vec{r} = \omega(t+\Delta t) - \vec{k}\cdot(\vec{r}+\Delta\vec{r}) = const.$

$\Rightarrow \quad \omega\Delta t - \vec{k}\cdot\Delta\vec{r} = 0, \quad \frac{\Delta\vec{r}}{\Delta t} = \frac{\omega}{k^2}\vec{k} = \vec{v}_\phi$

能/动量.

$E = \hbar\omega. \quad \vec{p} = \hbar\vec{k}$

群速度.

$E_1 = E_0 \cos[(k+\Delta k)x - (\omega+\Delta\omega)t]$

$E_2 = E_0 \cos[(k-\Delta k)x - (\omega-\Delta\omega)t]$.

令 $a = kx - \omega t$. $b = (\Delta k)x - (\Delta\omega)t$.

$E_1 + E_2 = E_0[\cos(a+b) + \cos(a-b)] = 2E_0 \cos a \cos b$

$$E_1 + E_2 = 2E_0 \cos(\Delta k x - \Delta\omega t)\cos(kx - \omega t).$$

调制波、包络（低频）

包络的"相速度"：群速度.

$$v_g = \frac{\Delta\omega}{\Delta k} \sim \frac{d\omega}{dk}.$$

色散关系：

$$f(\omega, \vec{k}) = 0.$$

用色散关系，描述波的信息. 如何求解？

- 线性.
- 非线性.

Damping . instability . balance ?

$$E = E_0 \exp[i(\vec{k}\cdot\vec{r} - \omega t)].$$

Example : $\omega^4 = (c^2k^2 + \omega_p^2) + \omega_c^4$.

(1) ω is real. $\vec{k} = \vec{k}_r + i\vec{k}_i$ 空间阻尼 or 空间不稳定性.

$$E = E_0 \exp[i(\vec{k}\cdot\vec{r} - \omega t)] = E_0 e^{i(\vec{k}_r + i\vec{k}_i)\cdot\vec{r}} e^{-i\omega t}.$$

$$= E_0 e^{-\vec{k_i}\cdot\vec{r}} e^{i(\vec{k}\cdot\vec{r}-\omega t)}$$

E_0 振幅在空间中衰减.

if $\vec{k_i}\cdot\vec{r} > 0$. $E_0 \downarrow$.

$\vec{k_i}\cdot\vec{r} < 0$. $E_0 \uparrow$

$\vec{k_i}\cdot\vec{r} = 0$ E_0 remain

(2). $\vec{k}$ is real. $\omega = \omega_r + i\omega_i$

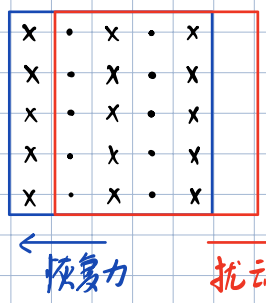
$$\vec{E} = \vec{E_0} e^{i(\vec{k}\cdot\vec{r}-\omega t)} = \underbrace{\vec{E_0} e^{\omega_i t}}_{\vec{E}} e^{i(\vec{k}\cdot\vec{r}-\omega_r t)}$$

$\omega_i > 0$ instability

$\omega_i < 0$ damping

$\omega_i = 0$ ~

均匀等离子体.



分离场 $\nabla\cdot\vec{E}$

•：电子

x：离子.

cold ion assumption：

$n_i = n_0$.

$$\rho = q_i n_i - q_e n_e = (n_0 - n_e)e$$

$$\nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0} \qquad \vec{B} = 0.$$

连续性方程：

$$\frac{\partial n_e}{\partial t} + \nabla\cdot(n_e\vec{v_e}) = 0$$

动力学方程.

$$m d\vec{v_e} = -ed\vec{E} \Rightarrow \frac{\partial\vec{v_e}}{\partial t} + (\vec{v_e}\cdot\nabla)\vec{v_e}$$

① 一维假设：

$$\begin{cases} \dfrac{\partial \vec{E}_x}{\partial x} = \dfrac{(n_0 - n_e)}{\varepsilon_e} & (a) \\ \dfrac{\partial n_e}{\partial t} + \dfrac{\partial (n_e v_e)}{\partial x} = 0 & (b) \\ m_e \dfrac{\partial \vec{v_e}}{\partial t} + m_e v_e \dfrac{\partial v_e}{\partial x} = -eE_x & (c) \end{cases}$$

② wave assumption + 线性微扰.

微扰：　E：无静电场.　$E_x = \underbrace{E_{x0}}_{=0} + E_{x1}$

n_e: $n_e = \underbrace{n_{e0}}_{=n_0} + n_{e1}$

v_e: $v_e = \underbrace{v_{e0}}_{=0} + v_{e1}$

$E_x,\ n_e,\ v_e \propto \exp[i(kx-\omega t)]$.

$\therefore$ $$\frac{\partial E_{x1}}{\partial t} = \frac{e(n_0 - n_0 - n_{e1})}{\epsilon} = -\frac{e n_{e1}}{\epsilon} \qquad (a)$$

$$\frac{\partial}{\partial t}(n_{e1}) + \frac{\partial}{\partial x}[\underbrace{(n_{e0} + n_{e1})v_{ex1}}_{n_{e0}v_{ex1} + \cancel{n_{e1}v_{ex1}}}] = 0.$$

二阶无穷小.

$$\frac{\partial n_{e1}}{\partial t} + \frac{\partial}{\partial x}(n_{e0}v_{ex1}) = 0. \qquad (b)$$

$$m_e\left(\frac{\partial v_{ex1}}{\partial t} + v_{ex1}\cancel{\frac{\partial}{\partial x}v_{ex1}}\right) = -eE_{x1} \qquad (c)$$

二阶无穷小.

$$\left.\begin{array}{ll} : & ikE_{x1} = -\dfrac{e n_e}{\epsilon} \\ : & -i\omega n_{e1} + ikn_0 v_{ex1} = 0 \\ : & -i\omega m_e v_{ex1} = -eE_{x1} \end{array}\right\}$$

有非零解: $|A| = 0$ —— $f(\omega, k) = 0$.

$$\omega_{pe}^2 = \frac{n_0 e^2}{\varepsilon_0 m_e}$$